HDOC – 17

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PROBLEM STATEMENT:

Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer. Define one separate user-defined function for each menu option, *with void return type* and *with a simple function call (without call by value/reference)*. Each function should read the required input, process it, and display the result. Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

- Generate the largest possible number using its digits
- Generate the smallest possible number using its digits.
- Swap its first and last digits.
- Check whether it is a palindrome number.

Algorithm:

1. Start the program.
2. Declare four user-defined functions:
    - largest() – generates the largest number using the digits.
    - smallest() – generates the smallest number using the digits.
    - swapDigits() – swaps the first and last digits.
    - palindrome() – checks whether the number is a palindrome.
3. Display the menu:
    - 1 → Largest possible number
    - 2 → Smallest possible number

   - 3 → Swap first and last digits
   - 4 → Check palindrome
4. Read the user's choice.
5. Use switch(choice):
   - If choice is 1, call largest().
   - If choice is 2, call smallest().
   - If choice is 3, call swapDigits().
   - If choice is 4, call palindrome().
   - Otherwise, display "Invalid choice".
6. Each function takes the positive integer from the user, performs its operation, and displays the result.
7. Stop.

Test Case table:

Test case	Choice	Input	Expected output	Actual output	Result
1.	1	12054	Largest number = 54210	Largest number = 54210	Pass
2.	2	4021	Smallest number = 1024	Smallest number = 1024	pass
3.	3	9876	Number after swapping = 6879	Number after swapping = 6879	pass
4.	4	12321	12321 is a palindrome number	12321 is a palindrome number	pass
5.	5	_______	Invalid choice	Invalid choice	pass

C Program:

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1 2 3 4

Open

```
    return 0;
}


void largest()
{
    int n, digit, freq[10] = {0};

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while(n > 0)
    {
        digit = n % 10;
        freq[digit]++;
        n = n / 10;
    }

    printf("Largest number = ");

    for(int i = 9; i >= 0; i--)
    {
        while(freq[i] > 0)
        {
            printf("%d", i);
            freq[i]--;
        }
    }

    printf("\n");
}


void smallest()
{
    int n, digit, freq[10] = {0};

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while(n > 0)
    {
        digit = n % 10;
        freq[digit]++;
        n = n / 10;
    }
```

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1 2 3 4

Open

```c

    while(n > 0)
    {
        digit = n % 10;
        freq[digit]++;
        n = n / 10;
    }

    printf("Smallest number = ");

    for(int i = 1; i <= 9; i++)
    {
        if(freq[i] > 0)
        {
            printf("%d", i);
            freq[i]--;
            break;
        }
    }

    for(int i = 0; i <= 9; i++)
    {
        while(freq[i] > 0)
        {
            printf("%d", i);
            freq[i]--;
        }
    }

    printf("\n");
}


void swapDigits()
{
    int n, first, last, temp, power = 1, result;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    last = n % 10;
    temp = n;

    while(temp >= 10)
    {
```

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1 2 3 4

Open

```c
void swapDigits()
{
    int n, first, last, temp, power = 1, result;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    last = n % 10;
    temp = n;

    while(temp >= 10)
    {
        temp = temp / 10;
        power = power * 10;
    }

    first = temp;

    result = n - first * power - last;
    result = result + last * power + first;

    printf("Number after swapping = %d\n", result);
}


void palindrome()
{
    int n, temp, digit, reverse = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;

    while(temp > 0)
    {
        digit = temp % 10;
        reverse = reverse * 10 + digit;
        temp = temp / 10;
    }

    if(n == reverse)
        printf("%d is a palindrome number.\n", n);
    else
        printf("%d is not a palindrome number.\n", n);
}
```

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File Machine View Input Devices Help

1 2 3 4

Open

```c
#include <stdio.h>

void largest();
void smallest();
void swapDigits();
void palindrome();

int main()
{
    int choice;

    printf("----- MENU -----\n");
    printf("1. Generate largest number\n");
    printf("2. Generate smallest number\n");
    printf("3. Swap first and last digits\n");
    printf("4. Check palindrome\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            largest();
            break;

        case 2:
            smallest();
            break;

        case 3:
            swapDigits();
            break;

        case 4:
            palindrome();
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

Compile the programing using:

gcc hdoc17.c -o hdoc17

Execute the code using:

./hdoc17

Here are some screenshots of compilation and execution against test case: